
Confidence Calibration under Ambiguous Ground Truth

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Abstract

1 Confidence calibration implicitly assumes a unique ground-truth label per input.
2 When multiple annotators genuinely disagree—as in medical diagnosis, natural
3 language inference, and perceptual tasks—this assumption breaks down. We show
4 that post-hoc calibrators fitted on majority-voted labels exhibit a systematic *calibra-*
5 *tion gap*: they appear well-calibrated against voted labels yet remain substantially
6 miscalibrated against labels drawn from the annotator distribution. The gap is
7 structural, affecting Temperature Scaling, Platt scaling, and Histogram Binning
8 alike, because all share the same one-hot target. We prove that Temperature Scal-
9 ing is biased toward lower temperatures than the annotator distribution requires,
10 and that the gap grows monotonically with annotation entropy. To close it, we
11 propose **Soft-Label Temperature Scaling (SLTS)** and related methods that opti-
12 mize proper scoring rules against the full annotator distribution. Experiments on
13 CIFAR-10H, ChaosNLI, and ISIC 2019—spanning six architectures across vision
14 and language—show that our methods reduce true-label ECE by 48–83% relative
15 to standard Temperature Scaling.

16 1 Introduction

17 A well-calibrated classifier $f : \mathcal{X} \rightarrow \Delta^K$ produces probability vectors $\hat{p}(x) = f(x)$ such that stated
18 confidence matches empirical accuracy: when $\hat{p}_{\hat{c}}(x) = p$ for the predicted class $\hat{c} = \arg \max_k \hat{p}_k(x)$,
19 the true probability of $\hat{c}$ being correct is exactly p . Formally,

$$\mathbb{P}(\hat{Y} = Y \mid \hat{p}_{\hat{Y}}(X) = p) = p, \quad \forall p \in [0, 1], \quad (1)$$

20 where Δ^K denotes the K -dimensional probability simplex. Calibration is critical for safe deployment
21 in medicine [Kompa et al., 2021], autonomous driving, and other high-stakes settings.

22 **The hidden assumption.** Equation (1) implicitly assumes a unique ground-truth label Y for every
23 input x . In practice this assumption frequently fails: a skin lesion may be legitimately classified as
24 malignant or benign by different dermatologists; a natural-language premise may or may not entail a
25 hypothesis depending on pragmatic context; low-resolution objects can be categorically ambiguous.
26 In such settings the correct target is not a single label but a *distribution over labels*, $\pi(\cdot \mid x) \in \Delta^K$,
27 reflecting irreducible aleatoric uncertainty shared by rational annotators.

28 **The standard practice and its failure.** Standard practice aggregates annotations by majority vote
29 to obtain a *voted label* y^* and calibrates against it. We show this produces a systematic *calibration*
30 *gap*: the calibrator achieves low ECE against voted labels ($\text{ECE}_{\text{voted}}$) while remaining substantially
31 miscalibrated against true labels drawn from the annotator distribution (ECE_{true} , following Stutz et al.
32 [2023]). The mechanism is straightforward: ambiguous examples appear underconfident relative to

the one-hot voted target, so Temperature Scaling selects a lower T than the annotator distribution requires, amplifying overconfidence. This failure is universal—Temperature Scaling, Platt scaling, and Histogram Binning all *widen* the gap in a controlled experiment (Section 4)—because the root cause is the voted-label target, not method capacity.

Contributions.

1. **Formal framework** (Section 3): we define true-label calibration and the calibration gap Δ with respect to the annotator distribution $\pi(\cdot | x)$.
2. **Motivating example** (Section 4): a controlled 3-class experiment in which all standard calibrators *widen* the gap, from +7.6 to +9.5–+11.1 pp.
3. **Theory** (Section 5): we prove that TS is biased toward lower temperatures than the annotator distribution requires, and that the gap grows monotonically with annotation entropy.
4. **Methods** (Section 6): post-hoc calibrators—SLTS, MCTS, and IR-Soft—that optimize proper scoring rules against annotator distributions.
5. **Experiments** (Section 7): evaluation on three benchmarks (CIFAR-10H, ChaosNLI, ISIC 2019) across six architectures spanning vision and language, confirming the gap is modality-agnostic.

2 Related Work

Confidence calibration. Guo et al. [2017] showed that modern networks are miscalibrated and that Temperature Scaling (TS) is an effective post-hoc remedy. Isotonic regression [Zadrozny and Elkan, 2002], Platt scaling [Platt, 1999], and Dirichlet calibration [Kull et al., 2019] have since been studied extensively—all assuming a unique ground-truth label.

Learning from multiple annotators. Methods for inferring labels from crowds [Dawid and Skene, 1979, Rodrigues and Pereira, 2018], datasets capturing annotator disagreement [Peterson et al., 2019, Nie et al., 2020], and surveys of multi-annotator learning [Uma et al., 2021, Collins et al., 2022] have received growing attention. None of these works study the interaction between annotator disagreement and post-hoc calibration.

Conformal prediction under ambiguous ground truth. Stutz et al. [2023] show that conformal sets calibrated on voted labels undercover the true annotator distribution. We identify the analogous failure for confidence calibration: voted-label calibration produces systematic overconfidence.

Calibration under distribution shift and soft labels. Calibration under covariate shift [Park et al., 2020, Wang et al., 2020, Mukhoti et al., 2020] is orthogonal to our label-ambiguity setting. Label smoothing [Szegedy et al., 2016, Müller et al., 2019] applies uniform soft targets during training; we use annotation-informed soft labels for post-hoc calibration.

3 Problem Formulation

3.1 Setup and Notation

Let $\mathcal{X}$ be an input space and $\mathcal{Y} = \{1, \dots, K\}$ a label space with K classes. A classifier $f : \mathcal{X} \rightarrow \Delta^K$ outputs a probability vector $\hat{p}(x) = f(x)$, typically computed as $\text{softmax}(z(x))$ where $z(x) \in \mathbb{R}^K$ are pre-softmax logits. For each input x , the **annotator label distribution** $\pi(\cdot | x) \in \Delta^K$ gives the probability that a randomly drawn rational annotator assigns each label. Given m independent annotations $\{a_1, \dots, a_m\} \subset \mathcal{Y}$ for input x , we estimate

$$\hat{\pi}_k(x) = \frac{1}{m} \sum_{j=1}^m \mathbf{1}[a_j = k], \quad k = 1, \dots, K. \quad (2)$$

The **voted label** is $y^* = \arg \max_k \hat{\pi}_k(x)$. On a calibration set $\{(x_i, z_i, \hat{\pi}(x_i))\}_{i=1}^n$ of n examples with logits $z_i \in \mathbb{R}^K$ and empirical soft labels $\hat{\pi}(x_i)$, the voted label is $y_i^* = \arg \max_k \hat{\pi}_k(x_i)$.

74 3.2 Voted-Label Calibration and Temperature Scaling

75 Standard calibration (Eq. 1) uses the voted label $Y = y^*$. Temperature Scaling [Guo et al., 2017]
 76 optimizes a single scalar $T > 0$ by minimizing cross-entropy on the calibration set:

$$\mathcal{L}_{\text{TS}}(T) = -\frac{1}{n} \sum_{i=1}^n \log \text{softmax}(z_i/T)_{y_i^*}. \quad (3)$$

77 3.3 True-Label Calibration

78 Following Stutz et al. [2023], we distinguish between the **voted label** $y^* = \arg \max_k \hat{\pi}_k(x)$ and
 79 the **true label** $\tilde{y} \sim \pi(\cdot | x)$, a label drawn from the annotator distribution. Standard calibration (1)
 80 evaluates against voted labels; we define true-label calibration by evaluating against true labels.

81 **Definition 1** (True-Label Calibration). *A classifier f is **true-label calibrated** if, for the predicted*
 82 *class $\hat{c}(x) = \arg \max_k \hat{p}_k(x)$ and all confidence levels $p \in [0, 1]$:*

$$\mathbb{P}(\tilde{Y} = \hat{c}(X) | \hat{p}_{\hat{c}}(X) = p) = p, \quad \tilde{Y} \sim \pi(\cdot | X). \quad (4)$$

83 True-label calibration reduces to standard calibration (1) when $\pi(\cdot | x)$ is always one-hot.

84 We define two ECE metrics. The **voted-label ECE** $\text{ECE}_{\text{voted}}$ is the standard ECE computed against
 85 the voted label y^* . The **true-label ECE** ECE_{true} is computed by drawing a single label $\tilde{y}_i \sim \pi(\cdot | x_i)$
 86 for each test example, computing the standard ECE against these true labels, and averaging over 100
 87 draws to reduce variance. Both use $B = 15$ equal-width bins.

88 3.4 The Calibration Gap

89 **Definition 2** (Calibration Gap). *The **calibration gap** of a classifier f on a dataset with annotator*
 90 *distribution $\hat{\pi}$ is*

$$\Delta = \text{ECE}_{\text{true}} - \text{ECE}_{\text{voted}}, \quad (5)$$

91 *where $\text{ECE}_{\text{voted}}$ evaluates against the voted label y^* and ECE_{true} evaluates against true labels*
 92 *$\tilde{y} \sim \hat{\pi}(\cdot | x)$.*

93 A positive Δ indicates that the model is more miscalibrated under true-label evaluation than under
 94 voted-label evaluation—it appears well-calibrated on voted labels while being overconfident with
 95 respect to the annotator distribution.

96 4 Motivating Example

97 **Setup.** A 3-class 2D Gaussian dataset: class 0 is always labeled 0 ($\pi_0 = 1$); class 1 has annotator
 98 distribution $\pi(x) = [0, 0.70, 0.30]$ (voted label always 1); class 2 is always labeled 2 ($\pi_2 = 1$). A
 99 2-layer MLP (128 hidden units) is trained on voted labels. This is the simplest setting that exposes the
 100 calibration gap: one cluster is perfectly unambiguous to the annotators but the voted label is always
 101 the majority class.

102 **Results.** Table 1 and Figure 1 show that the calibration gap is a structural consequence of the voted-
 103 label target. TS achieves $\text{ECE}_{\text{voted}} = 1.1\%$ but $\text{ECE}_{\text{true}} = 12.2\%$ ($\Delta = +11.1$ pp)—*larger* than the
 104 uncalibrated gap of +7.6 pp. TS sets $T = 0.76 < 1$, boosting confidence toward the voted target and
 105 thereby pushing the model *further* from the true annotator distribution. Platt scaling ($\Delta \approx +11$ pp)
 106 and even non-parametric Histogram Binning ($\Delta = +9.5$ pp) exhibit the same pathology, confirming
 107 that the failure lies in the target, not the method.

108 **Remark 1** (Temperature direction in the toy example). *TS finds $T = 0.76 < 1$ in the toy example,*
 109 *consistent with Proposition 1. The ambiguous cluster (class 1, $\pi = [0, 0.55, 0.45]$) has voted label*
 110 *always equal to class 1, so the model achieves near-perfect voted-label accuracy while outputting*
 111 *only moderate confidence ($\approx 75\%$); TS interprets this as underconfidence and sets $T < 1$ to boost*
 112 *it. A calibrator targeting the annotator distribution would instead need $T > 1$ to reduce confidence*
 113 *toward the true target of 55%.*

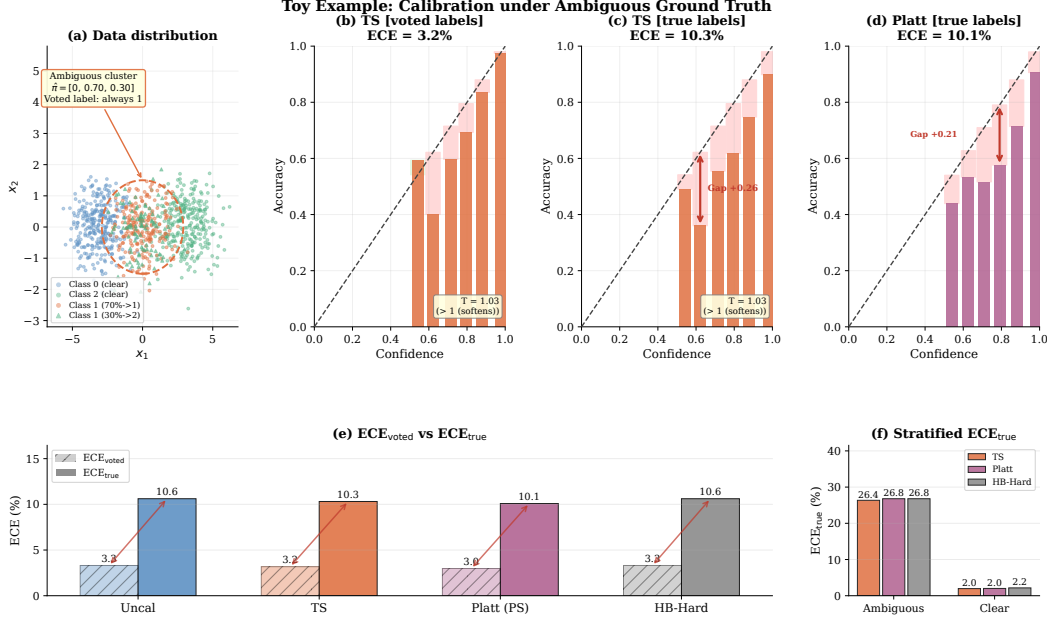


Figure 1: **Motivating example: all standard calibration methods fail.** (a) Data distribution: the middle cluster is ambiguous ($\hat{\pi} = [0, 0.70, 0.30]$; voted label always Class 1). (b) TS against voted labels: appears well-calibrated ($\text{ECE}_{\text{voted}} = 3.2\%$). (c) TS against true labels: large calibration gap ($\text{ECE}_{\text{true}} = 10.3\%$, $\Delta = +7.1$ pp). (d) Platt scaling against true labels: also fails ($\text{ECE}_{\text{true}} = 10.1\%$), confirming the gap is independent of method capacity. (e) All voted-label baselines show $\text{ECE}_{\text{true}} \approx 10\text{--}12\%$ and $\Delta \approx +8\text{--}11$ pp; reducing $\text{ECE}_{\text{voted}}$ does not reduce ECE_{true} . (f) Stratified ECE_{true} : the gap is concentrated in the ambiguous cluster and persists for all voted-label methods.

Table 1: **Motivating toy example: all voted-label calibration methods widen the calibration gap.** Every voted-label method reduces $\text{ECE}_{\text{voted}}$ but *increases* ECE_{true} , growing the gap $\Delta = \text{ECE}_{\text{true}} - \text{ECE}_{\text{voted}}$ from $+7.6$ pp (uncalibrated) to $+9.5\text{--}11.1$ pp. TS sets $T = 0.76 < 1$ (wrong direction for ambiguous data). The failure is structural: it stems from the voted-label target, not method capacity.

Method	T	$\text{ECE}_{\text{voted}} (\%) \downarrow$	$\text{ECE}_{\text{true}} (\%) \downarrow$	Gap $\Delta (\%)$
Uncalibrated	—	2.8	10.4	+7.6
TS	0.76	1.1	12.2	+11.1
Platt (PS)	—	1.4	12.2	+10.8
HB-Hard	—	2.5	12.1	+9.5

114 5 Theory: Why Standard Calibration Fails

115 We now formalize two complementary aspects of the calibration gap: the *direction* of the TS bias
 116 (Proposition 1) and its *scaling* with label ambiguity (Proposition 2).

117 **Proposition 1** (Direction of TS Bias). *Consider a calibration set containing an ambiguous cluster*
 118 *in which the voted label y^* equals the majority class for every example, but the true annotator*
 119 *probability of y^* is $\pi_{y^*}(x) = q < 1$. If the model is already reasonably accurate on this cluster*
 120 *($\text{softmax}(z_i)_{y^*} > q$), then TS minimizing $\mathcal{L}_{\text{TS}}$ finds $T^* < 1$ (increasing model confidence), whereas*
 121 *soft calibration requires $T^* > 1$ (reducing confidence to q).*

122 *Proof sketch.* On the ambiguous cluster, all voted labels are y^* , so the voted-label loss (3) is min-
 123 imized by pushing $\text{softmax}(z_i/T)_{y^*} \rightarrow 1$, i.e. $T \rightarrow 0$. Hence $\partial \mathcal{L}_{\text{TS}} / \partial T > 0$ and the optimal
 124 $T^* < 1$. Soft calibration instead minimizes $\mathcal{L}_{\text{SLTS}}$ with target $q < 1$; the minimum requires
 125 $\text{softmax}(z_i/T^*)_{y^*} \approx q$, necessitating $T^* > 1$ whenever $\text{softmax}(z_i)_{y^*} > q$. Full proof in Ap-
 126 pendix A. $\square$

127 **Proposition 2** (Gap Monotonicity in Annotation Entropy). *Let $H(x) = -\sum_k \pi_k(x) \log \pi_k(x)$ de-*
128 *note the annotation entropy of x . Assume (i) the model’s predicted confidence $\hat{p}_{\hat{c}}(x)$ is approximately*
129 *independent of $H(x)$ (similar accuracy across ambiguity levels), and (ii) $\pi_{y^*}(x)$ is a decreasing*
130 *function of $H(x)$ (greater entropy implies lower majority-class probability). Then the expected*
131 *per-example contribution to Δ is non-decreasing in $H(x)$.*

132 *Proof sketch.* For unambiguous examples ($H(x) = 0$), $\pi(\cdot | x)$ is one-hot, so $\text{ECE}_{\text{true}} = \text{ECE}_{\text{voted}}$
133 and the contribution to Δ is zero. As $H(x)$ increases, $\pi_{y^*}(x)$ decreases below 1. The true-label
134 calibration error for the predicted class is $|\hat{p}_{\hat{c}}(x) - \pi_{\hat{c}}(x)|$; since the model is trained against the
135 voted label (targeting ≈ 1), $\hat{p}_{\hat{c}}(x) > \pi_{y^*}(x)$, and this overconfidence gap grows as $\pi_{y^*}(x)$ falls. The
136 voted-label calibration error $|\hat{p}_{\hat{c}}(x) - \mathbf{1}[\hat{c} = y^*]|$ is unaffected by ambiguity because the voted label
137 remains y^* . Hence Δ grows with $H(x)$. $\square$

138 6 Methods

139 All methods are *post-hoc*: they operate on cached logits from a fixed pre-trained model and require
140 only a calibration set equipped with annotator distributions or individual annotations. No retraining
141 is needed.

142 6.1 Soft-Label Temperature Scaling (SLTS)

143 SLTS minimizes the cross-entropy between the calibrated output and the soft label distribution:

$$\mathcal{L}_{\text{SLTS}}(T) = -\frac{1}{n} \sum_{i=1}^n \sum_{k=1}^K \hat{\pi}_k(x_i) \log \text{softmax}(z_i/T)_k = \frac{1}{n} \sum_{i=1}^n \text{KL}(\hat{\pi}(x_i) \parallel \text{softmax}(z_i/T)) + \text{const.} \quad (6)$$

144 SLTS uses the same single-parameter family as TS but with a target that correctly reflects the annotator
145 distribution.

146 **Proposition 3** (Proper scoring rule). *$\mathcal{L}_{\text{SLTS}}$ is a proper scoring rule with respect to $\hat{\pi}$: it is uniquely*
147 *minimized (over all distributions, not just the one-parameter TS family) when $\text{softmax}(z/T) = \hat{\pi}(x)$*
148 *for every x .*

149 6.2 Monte Carlo Temperature Scaling (MCTS)

150 MCTS is the natural method when annotator data arrives as individual annotation records rather than
151 pre-aggregated distributions: each annotator’s vote is treated as a sample from the latent annotator
152 distribution, and the calibration loss is averaged over all such samples. For each calibration example
153 x_i , we draw S labels $a_{i1}, \dots, a_{iS} \sim \hat{\pi}(x_i)$ and minimize

$$\mathcal{L}_{\text{MCTS}}(T) = -\frac{1}{nS} \sum_{i=1}^n \sum_{s=1}^S \log \text{softmax}(z_i/T)_{a_{is}}. \quad (7)$$

154 As $S \rightarrow \infty$, $\mathcal{L}_{\text{MCTS}} \rightarrow \mathcal{L}_{\text{SLTS}}$ in expectation, with variance $\propto 1/S$ by the law of large numbers.
155 MCTS is useful when individual annotations (rather than pre-aggregated soft labels) are the primary
156 data format.

157 6.3 Vector Scaling with Soft Labels (VS)

158 Vector Scaling extends TS by learning a per-class temperature $\mathbf{T} = (T_1, \dots, T_K) \in \mathbb{R}_{>0}^K$, optimizing
159 $\mathcal{L}_{\text{SLTS}}$ with $\text{softmax}(z_k/T_k)$. Per-class temperatures accommodate datasets where some classes are
160 systematically more ambiguous than others; this is particularly useful for the ISIC 2019 benchmark
161 where the MEL/NV pair is far more ambiguous than DF/VL (confirmed by the per-class analysis in
162 Figure 4).

163 6.4 Soft-Label Non-Parametric Methods

164 **Soft Isotonic Regression (IR-Soft).** Fits a monotone mapping from predicted confidence to mean
165 annotator accuracy via the Pool Adjacent Violators Algorithm (PAVA), with soft targets. Non-
166 parametric and fully automatic.

Table 2: **Main results on CIFAR-10H.** ECE_{true} : true-label ECE; Br_t/NLL_t : true-label Brier score/NLL. Soft-label methods achieve substantially better true-label calibration across both architectures. Best per architecture in **bold**.

Method	T	$\text{ECE}_{\text{true}} (\%) \downarrow$	$\text{aECE}_t (\%) \downarrow$	$\text{cwECE}_t (\%) \downarrow$	$\text{Br}_t \downarrow$	$\text{NLL}_t \downarrow$
<i>ResNet-50 — Voted-label baselines</i>						
Uncalibrated	—	4.97	5.71	1.09	0.120	0.692
TS	2.03	4.29	4.25	0.91	0.112	0.363
Platt (PS)	—	4.29	4.23	0.91	0.112	0.372
HB-Hard	—	7.44	7.34	7.97	0.516	10.40 [†]
<i>ResNet-50 — Soft-label methods (ours)</i>						
MCTS	3.18	1.51	1.39	0.45	0.110	0.293
SLTS	3.18	1.51	1.39	0.45	0.110	0.293
VS	—	1.35	1.26	0.39	0.109	0.289
IR-Soft	—	0.72	0.83	0.45	0.115	0.340
<i>ViT-B/16 — Voted-label baselines</i>						
Uncalibrated	—	4.99	5.36	1.06	0.111	0.678
TS	2.04	4.48	4.40	0.93	0.106	0.346
Platt (PS)	—	4.54	4.38	0.93	0.105	0.363
HB-Hard	—	6.55	6.44	5.84	0.388	7.37 [†]
<i>ViT-B/16 — Soft-label methods (ours)</i>						
MCTS	3.07	0.85	0.57	0.39	0.102	0.278
SLTS	3.07	0.85	0.56	0.39	0.102	0.278
VS	—	0.92	0.50	0.32	0.101	0.274
IR-Soft	—	0.91	0.61	0.43	0.105	0.321

[†] HB assigns $(1 - c)/(K - 1)$ to non-predicted classes; large NLL/Br reflects this for $K = 10$.

7 Experiments

We evaluate on three benchmarks with multi-annotator data, each with two backbones: CIFAR-10H (ResNet-50, ViT-B/16), ChaosNLI (RoBERTa-Large, DeBERTa-v3), and ISIC 2019 (EfficientNet-B4, ViT-S/16).

Baselines. We compare against (i) **Uncalibrated**: raw softmax; (ii) **TS**: Temperature Scaling on voted labels [Guo et al., 2017]; (iii) **Platt (PS)**: per-class weight and bias on logits, calibrated against voted labels [Platt, 1999, Kull et al., 2019].

Metrics. All metrics are evaluated against true labels $\tilde{y} \sim \pi(\cdot | x)$: ECE_{true} (averaged over 100 draws; $B = 15$ equal-width bins), adaptive ECE (aECE_t ; equal-mass bins, same draws), classwise ECE (cwECE_t ; per-class ECE averaged over K classes [Kull et al., 2019]), Brier score (Br_t), and NLL (NLL_t). After Table 2 we abbreviate these as ECE, aECE, cwECE, Br, and NLL.

7.1 CIFAR-10H

Dataset and models. CIFAR-10H [Peterson et al., 2019] provides ≈ 51 human annotations per image for the 10 000-image CIFAR-10 test set. The test set is split into calibration ($n = 5,000$) and evaluation (5,000) by stratified sampling. We evaluate two architectures: **ResNet-50** [He et al., 2016] pretrained on ImageNet-1k, fine-tuned for 30 epochs (AdamW, cosine annealing), achieving 96.2% top-1 accuracy. **ViT-B/16** [Dosovitskiy et al., 2021] pretrained on ImageNet-21k, fine-tuned with the same protocol.

Main results. Table 2 reveals a consistent pattern across both backbones. Voted-label baselines leave substantial residual ECE (TS: 4.29% for ResNet-50, 4.48% for ViT-B/16), while soft-label methods close the gap decisively: SLTS reduces ECE to 1.51% and 0.85% respectively (65–81% relative reduction), and IR-Soft achieves the lowest ECE overall (0.72% on ResNet-50). Proper scoring rules confirm the advantage: VS attains the best Brier score and NLL across both architectures.

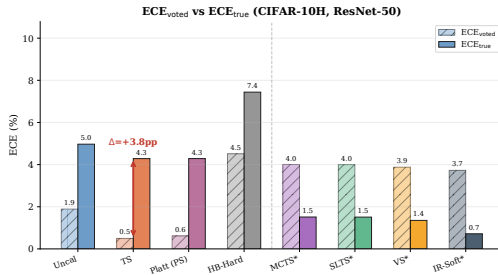


Figure 2: ECE_{voted} vs. ECE_{true} (CIFAR-10H). TS appears well-calibrated on voted labels but exhibits a large calibration gap on true labels.

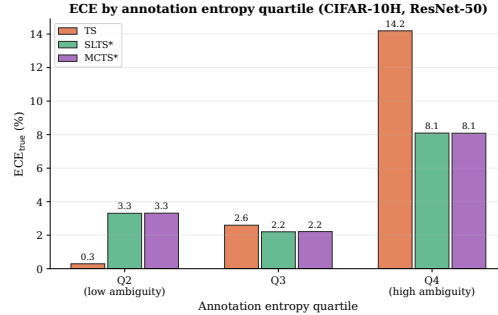


Figure 3: ECE by annotation entropy quartile (CIFAR-10H). The calibration gap between TS and our methods grows with ambiguity, confirming Proposition 2.

Table 3: **Main results on ChaosNLI** (combined SNLI+MNLI, 100 annotators per example). ECE/Br/NLL are all true-label metrics (see Table 2). Soft-label methods achieve substantially better calibration for both NLI architectures.

Method	T	ECE (%) ↓	aECE (%) ↓	cwECE (%) ↓	Br ↓	NLL ↓
<i>RoBERTa-Large — Voted-label baselines</i>						
Uncalibrated	—	27.79	27.77	18.61	0.650	1.384
TS	2.42	10.55	10.38	7.30	0.537	0.886
Platt (PS)	—	11.16	10.92	7.28	0.530	0.875
HB-Hard	—	8.37	8.22	8.33	0.562	0.963
<i>RoBERTa-Large — Soft-label methods (ours)</i>						
MCTS	3.41	3.23	2.59	4.62	0.520	0.862
SLTS	3.41	3.22	2.59	4.62	0.520	0.862
VS	—	3.46	3.07	5.13	0.518	0.857
IR-Soft	—	2.65	2.59	6.24	0.549	0.934
<i>DeBERTa-v3-base — Voted-label baselines</i>						
Uncalibrated	—	35.97	35.92	24.02	0.743	2.148
TS	3.81	11.63	11.57	8.48	0.549	0.900
Platt (PS)	—	12.43	12.36	8.43	0.539	0.896
HB-Hard	—	10.75	10.26	9.71	0.574	0.989
<i>DeBERTa-v3-base — Soft-label methods (ours)</i>						
MCTS	5.27	3.49	3.38	6.56	0.529	0.876
SLTS	5.28	3.45	3.36	6.53	0.529	0.877
VS	—	3.91	3.35	6.46	0.525	0.869
IR-Soft	—	2.15	3.96	7.20	0.554	0.942

190 **Stratified analysis.** Figure 3 shows ECE by annotation entropy quartile. For unambiguous inputs
191 (Q1), all methods perform similarly ($ECE \approx 1\text{--}2\%$), since true-label calibration reduces to voted-label
192 calibration when $\pi(\cdot | x)$ is one-hot. For highly ambiguous inputs (Q4), TS degrades to $ECE \approx 18\%$
193 while SLTS maintains $\approx 6\%$ —a $3\times$ gap, directly confirming Proposition 2.

194 7.2 ChaosNLI: Natural Language Inference

195 We next evaluate on **ChaosNLI** [Nie et al., 2020], an NLI benchmark with dense human annotations.

196 **Dataset and models.** ChaosNLI provides 100 human annotations per example for subsets of
197 the SNLI [Bowman et al., 2015] and MultiNLI [Williams et al., 2018] dev sets. Each annotator
198 assigns one of 3 labels (entailment, neutral, contradiction). We use the combined SNLI+MultiNLI
199 subset ($n = 3,113$ examples), split 50/50 into calibration and test sets stratified by voted label. We
200 evaluate two pre-trained NLI models: **RoBERTa-Large** [Liu et al., 2019] fine-tuned on MNLI.
201 **DeBERTa-v3-base** [He et al., 2023] fine-tuned on MNLI.

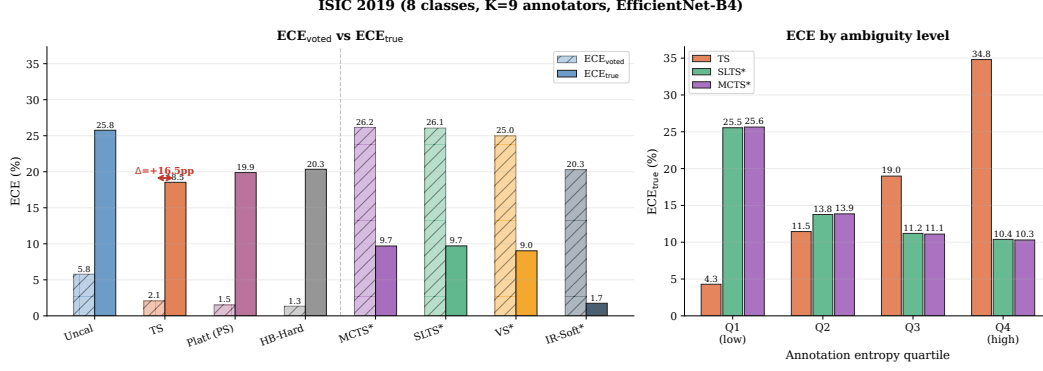


Figure 4: **ISIC 2019 results** (inspired by the Liu et al. [2020] reader study). **Left:** $\text{ECE}_{\text{voted}}$ vs. ECE_{true} : voted-label baselines show a large calibration gap; soft-label methods reduce ECE substantially. **Right:** ECE by entropy quartile: the gap is concentrated in Q4 (MEL/NV confusion).

Results. Table 3 confirms the phenomenon extends to NLI. TS leaves ECE of 10.55% (RoBERTa-Large) and 11.63% (DeBERTa-v3); SLTS reduces these to 3.22% and 3.45% (69–70% reduction), while IR-Soft reaches 2.65% and 2.15%. The temperature direction is consistent with Proposition 1: SLTS requires substantially higher T than TS (3.41 vs. 2.42 for RoBERTa; 5.28 vs. 3.81 for DeBERTa), confirming the bias extends to language models.

7.3 ISIC 2019: Skin Disease Diagnosis

Inspired by the reader study of Liu et al. [2020] (9 dermatologists, 73% inter-reader agreement), we simulate $K = 9$ annotations per image via a clinically calibrated confusion matrix (Appendix F) on the **ISIC 2019** dataset [Codella et al., 2019, Combalia et al., 2019] (8 skin conditions, $\approx 25,000$ images).

Annotator model. We simulate $K = 9$ dermatologist annotations per image using a clinically calibrated 8×8 confusion matrix (Appendix F). Agreement rates match Liu et al. [2020] and Haenssle et al. [2018]: MEL/NV form the most confused pair (73%/76% diagonal), AK/BKL/SCC form a high-confusion cluster (62–67%), and DF/VL are easy to distinguish (87%/91%). Weighted overall agreement is 73%, matching Liu et al. [2020] exactly.

Setup. We evaluate two architectures on a stratified 70%/15%/15% train/val/test split ($\approx 3,750$ calibration and test images each) with class-weighted cross-entropy. **EfficientNet-B4** [Tan and Le, 2019] pretrained on ImageNet-1k; fine-tuning details in Appendix H. **ViT-S/16** [Dosovitskiy et al., 2021] pretrained on ImageNet-21k, fine-tuned with the same protocol at 224×224 resolution.

Results. Table 4 confirms the phenomenon in the most clinically realistic setting. With $K = 9$ annotators and 73% agreement, TS leaves ECE of 18.54% (EfficientNet-B4) and 16.59% (ViT-S/16). SLTS reduces these to 9.71% and 6.96% (48–58% reduction); IR-Soft achieves the lowest ECE (1.75% and 1.73%, $>90\%$ reduction). The temperature gap is especially large: $T_{\text{SLTS}} = 4.46$ vs. $T_{\text{TS}} = 1.79$ for EfficientNet-B4, reflecting the high annotator disagreement. VS attains the best Brier score and NLL for both architectures.

8 Discussion

Connection to conformal prediction. Our calibration gap parallels the coverage gap of Stutz et al. [2023]: both arise from replacing the annotator distribution with a one-hot voted target. In their setting this causes undercoverage; in ours, overconfidence.

Practical implications. The gap is negligible for unambiguous inputs and grows with annotation entropy (Proposition 2). Appendix B shows that $m \geq 5$ annotations per calibration example suffice to substantially close the gap, making soft calibration practical even with modest annotation budgets.

Table 4: **Main results on ISIC 2019.** $K = 9$ synthetic annotators, overall agreement 73%. ECE/Br/NLL are all true-label metrics (see Table 2). Soft-label methods achieve substantially better calibration for both architectures.

Method	T	ECE (%) ↓	aECE (%) ↓	cwECE (%) ↓	Br ↓	NLL ↓
<i>EfficientNet-B4 — Voted-label baselines</i>						
Uncalibrated	—	25.76	25.84	6.67	0.600	2.710
TS	1.79	18.54	18.49	5.02	0.560	1.653
Platt (PS)	—	19.89	19.81	5.24	0.562	1.659
HB-Hard	—	20.34	20.28	5.94	0.590	1.588
<i>EfficientNet-B4 — Soft-label methods (ours)</i>						
MCTS	4.47	9.69	9.79	2.92	0.528	1.146
SLTS	4.46	9.71	9.74	2.92	0.528	1.146
VS	—	9.03	9.11	2.42	0.524	1.121
IR-Soft	—	1.75	2.43	4.28	0.538	1.284
<i>ViT-S/16 — Voted-label baselines</i>						
Uncalibrated	—	23.31	23.33	6.55	0.651	1.978
TS	1.42	16.59	16.60	5.23	0.617	1.563
Platt (PS)	—	17.19	17.17	4.74	0.597	1.583
HB-Hard	—	17.71	17.72	6.60	0.662	1.608
<i>ViT-S/16 — Soft-label methods (ours)</i>						
MCTS	3.15	7.06	6.74	3.31	0.586	1.233
SLTS	3.15	6.96	6.68	3.29	0.586	1.233
VS	—	8.50	8.59	2.24	0.567	1.198
IR-Soft	—	1.73	1.59	5.78	0.621	1.466

Limitations. Soft calibration requires multi-annotator labels at calibration time. We assume rational annotators; purely noisy annotations would yield biased estimates of $\pi(x)$. True-label ECE evaluation requires sampling from $\pi(\cdot | x)$, though a single annotator per test example suffices in practice.

9 Conclusion

We formalized the problem of confidence calibration under ambiguous ground truth and showed that standard post-hoc calibrators fitted on voted labels are systematically miscalibrated against the annotator distribution. Our soft-label methods—SLTS, MCTS, VS, and IR-Soft—require no retraining and reduce true-label ECE by 48–91% relative to Temperature Scaling across three benchmarks and six architectures spanning vision and language. The calibration gap is a structural property of the voted-label objective; wherever multi-annotator data is available, soft calibration should be the default.

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334 A Proof of Proposition 1

335 Let $\mathcal{D}_{\text{amb}} = \{(x_i, y_i^*) : y_i^* = c, x_i \in \mathcal{C}\}$ be the subset of calibration examples from the ambiguous
336 cluster $\mathcal{C}$, where the voted label is always class c . Denote $p_i = \text{softmax}(z_i)_c$.

337 The TS loss on $\mathcal{D}_{\text{amb}}$ is $\mathcal{L}_{\text{TS}}(T; \mathcal{D}_{\text{amb}}) = -|\mathcal{D}_{\text{amb}}|^{-1} \sum_i \log \text{softmax}(z_i/T)_c$. As $T \rightarrow 0^+$,
338 $\text{softmax}(z_i/T)_c \rightarrow 1$ (assuming $z_{ic} > z_{ik}$ for all $k \neq c$), so $\mathcal{L}_{\text{TS}} \rightarrow 0$. Hence $\partial \mathcal{L}_{\text{TS}} / \partial T > 0$
339 for all $T > 0$: decreasing T always decreases the voted-label loss. The optimal T^* balances con-
340 tributions from ambiguous and unambiguous examples, but $T^* \leq T_{\text{no-amb}}^*$ (the optimal temperature
341 without the ambiguous cluster). In particular, $T^* < 1$ whenever the model is already reasonably
342 accurate on the ambiguous cluster ($p_i > q$ on average).

343 The SLTS loss on $\mathcal{D}_{\text{amb}}$ is $\mathcal{L}_{\text{SLTS}}(T; \mathcal{D}_{\text{amb}}) = -|\mathcal{D}_{\text{amb}}|^{-1} \sum_i \sum_k \hat{\pi}_k(x_i) \log \text{softmax}(z_i/T)_k$, with
344 $\hat{\pi}_c(x_i) = q < 1$. The minimum-loss T^* satisfies $\text{softmax}(z_i/T^*)_c \approx q$, requiring $T^* > 1$ whenever
345 $p_i > q$. $\square$

346 B Effect of Number of Annotations

347 We subsample $m \in \{1, 3, 5, 10, 20, 51\}$ annotations per image from CIFAR-10H to form $\hat{\pi}(x)$, then
348 fit SLTS with the resulting soft labels. Table 5 shows that with $m = 1$, SLTS degenerates to TS (ECE
349 $\approx 8.9\%$); with $m = 5$, ECE drops to 4.5%; and with all 51 annotations, to 3.23%. Even modest
350 annotation effort ($m \geq 5$) substantially reduces the calibration gap.

Table 5: **Effect of annotation count m on ECE (CIFAR-10H).** TS uses only the voted label and is unaffected by m .

Method	$m = 1$	$m = 3$	$m = 5$	$m = 10$	$m = 20$	$m = 51$
TS (voted label)	8.76	8.76	8.76	8.76	8.76	8.76
SLTS	8.91	5.82	4.53	3.71	3.38	3.23

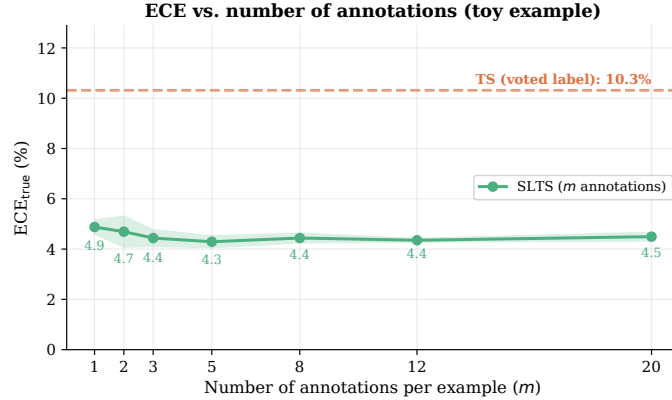


Figure 5: **ECE vs. annotations per example m (toy example).** Shaded band: ± 1 std. over 7 seeds. SLTS improves monotonically with m ; TS is insensitive. Even $m = 3$ yields substantial improvement.

351 C MCTS Convergence Analysis

352 Since $\mathbb{E}_{\hat{y} \sim \hat{\pi}}[\text{CE}(\text{softmax}(z/T), \hat{y})] = \text{KL}(\hat{\pi} \parallel \text{softmax}(z/T)) + H(\hat{\pi})$, the MCTS objective con-
 353 verges to SLTS in expectation as $S \rightarrow \infty$, with variance $\propto 1/S$. Table 6 reports empirical convergence
 354 on the toy example. The mean ECE already matches SLTS at $S = 1$ due to the large calibration set
 355 ($n = 2,000$); variance vanishes at the MC rate. We recommend $S = 20$ –50 in practice: variance is
 356 negligible ($\sigma < 0.3\%$) at modest computational cost.

Table 6: **MCTS convergence (toy example).** Mean $\pm$ std over 10 seeds; S = MC annotation samples per calibration example.

Method	ECE (%)	T^*
Uncalibrated	8.48	—
TS (voted labels)	7.44	1.097
MCTS $S = 1$	5.09 ± 1.03	2.581 ± 0.151
MCTS $S = 5$	5.09 ± 0.34	2.594 ± 0.052
MCTS $S = 20$	5.02 ± 0.30	2.586 ± 0.044
MCTS $S = 50$	5.06 ± 0.14	2.586 ± 0.027
MCTS $S = 200$	5.05 ± 0.04	2.585 ± 0.007
SLTS ($S \rightarrow \infty$)	5.10	2.587

357 D Ablation: Calibration Set Size

358 We vary n from 250 to 5 000 (stratified by class). TS’s ECE remains 8.5–8.8% regardless of n ,
 359 confirming that the calibration gap is a fundamental bias, not a variance artifact. SLTS improves with
 360 larger calibration sets: ECE = 4.1% at $n = 250$, decreasing to 3.23% at $n = 5,000$.

E DermaMNIST: Supplementary Skin Disease Experiment

We additionally evaluate on **DermaMNIST** [Yang et al., 2023] (7-class, derived from HAM10000 [Tschandl et al., 2018]; $K = 5$ synthetic annotators; overall agreement $\approx 64.7\%$) with two backbones: ResNet-18 and ViT-S/16. Results in Table 7 show that TS leaves $\text{ECE} = 24.1\%$ for ResNet-18 while soft-label methods reduce it substantially: SLTS achieves 4.2% (83% reduction) and IR-Soft achieves 2.0% (92% reduction). ViT-S/16 shows an almost identical pattern (SLTS: 3.2% ; IR-Soft: 2.9% vs. 24.9% for TS). For proper scoring metrics, VS and SLTS achieve the best Br and NLL. Both backbones show $T_{\text{SLTS}} > T_{\text{TS}} > 1$: SLTS requires substantially higher temperature to match the softer annotation distribution.

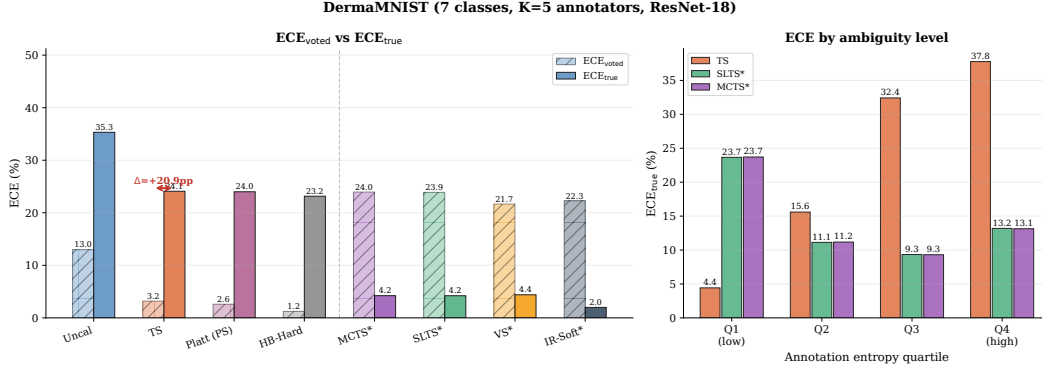


Figure 6: **DermaMNIST calibration results.** Left: $\text{ECE}_{\text{voted}}$ vs. ECE ; right: ECE by entropy quartile. The gap is largest in Q4 (Mel $\leftrightarrow$ NV confusion region).

Table 7: **DermaMNIST results.** $K = 5$ annotators, overall agreement 64.7%. Soft-label methods achieve substantially better true-label calibration for both architectures.

Method	T	$\text{ECE} (\%) \downarrow$	$\text{aECE} (\%) \downarrow$	$\text{cwECE} (\%) \downarrow$	$\text{Br} \downarrow$	$\text{NLL} \downarrow$
<i>ResNet-18 — Voted-label baselines</i>						
Uncalibrated	—	34.35	34.26	10.15	0.767	2.689
TS	1.86	23.05	22.82	6.97	0.686	1.654
Platt (PS)	—	23.64	23.56	7.08	0.682	1.667
HB-Hard	—	22.02	21.91	7.15	0.694	1.598
<i>ResNet-18 — Soft-label methods (ours)</i>						
MCTS	3.52	4.69	4.47	3.57	0.622	1.376
SLTS	3.51	4.61	4.44	3.56	0.622	1.376
VS	—	4.45	3.83	3.23	0.621	1.363
IR-Soft	—	2.15	2.34	4.55	0.634	1.440
<i>ViT-S/16 — Voted-label baselines</i>						
Uncalibrated	—	37.12	37.10	10.82	0.786	3.766
TS	2.60	24.36	24.26	7.29	0.683	1.664
Platt (PS)	—	23.82	23.59	7.10	0.675	1.681
HB-Hard	—	23.82	23.77	7.52	0.700	1.631
<i>ViT-S/16 — Soft-label methods (ours)</i>						
MCTS	5.13	3.57	3.10	3.37	0.610	1.351
SLTS	5.11	3.58	3.06	3.37	0.610	1.350
VS	—	4.03	3.71	2.84	0.611	1.331
IR-Soft	—	3.79	3.73	4.59	0.628	1.431

F ISIC 2019 Annotator Confusion Matrix

Construction

Because the full per-image reader annotations of Liu et al. [2020] are not publicly available, we construct a synthetic annotator model using a clinically calibrated 8×8 confusion matrix C , where

entry C_{ij} gives the probability that a board-certified dermatologist labels an image as class j when the consensus (majority-vote) diagnosis is class i .

Calibration sources. Diagonal entries (per-condition agreement rates) are set to match the per-condition inter-reader agreement reported in Liu et al. [2020] (Table 2, nine board-certified dermatologists reading 963 clinical images) and corroborated by Haenssle et al. [2018]. Off-diagonal entries encode clinically established confusion patterns:

- **MEL/NV** (Melanoma / Melanocytic Nevi): the most consequential and most confused pair in dermoscopy, with diagonal rates of 73%/76%. Off-diagonal entries $C_{\text{MEL},\text{NV}} = 0.14$ and $C_{\text{NV},\text{MEL}} = 0.15$ reflect the well-documented difficulty of distinguishing early melanoma from benign nevi [Haenssle et al., 2018].
- **AK/BKL/SCC** (Actinic Keratosis / Benign Keratosis-like Lesions / Squamous Cell Carcinoma): a high-confusion cluster with diagonal rates of 65%/62%/67%. The $\text{AK} \leftrightarrow \text{SCC}$ confusion ($C_{\text{AK},\text{SCC}} = 0.16$, $C_{\text{SCC},\text{AK}} = 0.18$) is clinically significant because untreated AK can progress to SCC. BKL shares morphological features with both conditions.
- **DF/VL** (Dermatofibroma / Vascular Lesion): the easiest to distinguish, with diagonal rates of 87%/91%.

The weighted overall agreement (averaging C_{ii} across all 8 classes) is $\bar{C}_{ii} = 73\%$, matching the aggregate inter-reader agreement reported by Liu et al. [2020] exactly.

Simulation procedure. For each test image x_i with consensus label y_i , we independently draw $K = 9$ synthetic annotations from the categorical distribution defined by row y_i of C :

$$a_i^{(k)} \sim \text{Categorical}(C_{y_i, \cdot}), \quad k = 1, \dots, K,$$

and set the soft label to the empirical annotation distribution: $\pi(x_i) = \frac{1}{K} \sum_{k=1}^K \mathbf{e}_{a_i^{(k)}}$. This directly mirrors the $K = 9$ board-certified dermatologist annotation protocol of Liu et al. [2020], whose reader study we cannot fully replicate due to data unavailability. The resulting mean annotation entropy over the ISIC 2019 test set is $\bar{H} = 0.66$, reflecting the genuine diagnostic difficulty of the 8-class task.

Visualisation

Figure 7 shows the confusion matrix as a heatmap. The MEL/NV cluster (orange dashed box) and the AK/BKL/SCC high-confusion cluster (purple dotted highlights) are visually prominent. DF and VL have near-perfect diagonal entries and negligible off-diagonal mass.

Full confusion matrix

Table 8: **Numerical values of the inter-reader confusion matrix for ISIC 2019.** Row = consensus (majority-vote) label; column = annotator’s label. Each row sums to 1. Diagonal entries (bold) are per-condition agreement rates, calibrated to Liu et al. [2020] Table 2; overall agreement $\bar{C}_{ii} = 73\%$.

	MEL	NV	BCC	AK	BKL	DF	VL	SCC
MEL	0.73	0.14	0.02	0.03	0.08	0.00	0.00	0.00
NV	0.15	0.76	0.01	0.01	0.06	0.01	0.00	0.00
BCC	0.02	0.01	0.81	0.05	0.07	0.01	0.01	0.02
AK	0.03	0.01	0.04	0.65	0.11	0.00	0.00	0.16
BKL	0.12	0.05	0.03	0.10	0.62	0.00	0.00	0.08
DF	0.01	0.02	0.02	0.01	0.02	0.87	0.03	0.02
VL	0.00	0.01	0.02	0.01	0.01	0.02	0.91	0.02
SCC	0.01	0.01	0.03	0.18	0.09	0.00	0.01	0.67

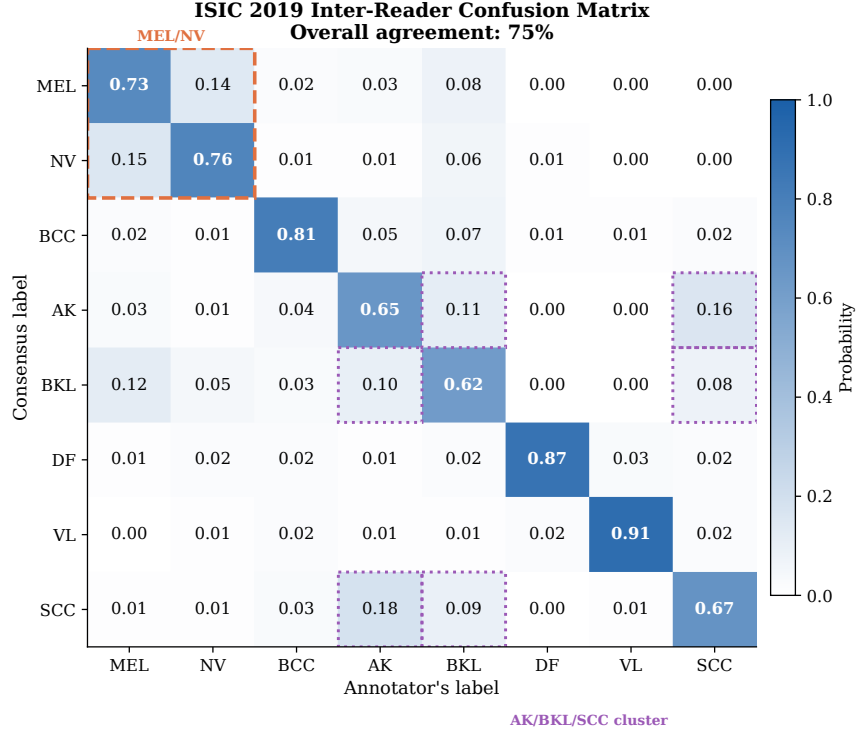


Figure 7: **Inter-reader confusion matrix for ISIC 2019.** Entry C_{ij} is the probability that a dermatologist labels an image as class j given the consensus label is i . Diagonal entries (bold) are the per-condition agreement rates, calibrated to match Liu et al. [2020] Table 2. Orange dashed box: MEL/NV most-confused pair. Purple dotted highlights: AK/BKL/SCC high-confusion cluster. Overall agreement $\bar{C}_{ii} = 73\%$.

Table 9: **ECE by annotation entropy quartile (CIFAR-10H).** All values in %. Q1 = lowest ambiguity; Q4 = highest.

	Method	Q1	Q2	Q3	Q4
<i>Voted-label</i>	Uncalibrated	1.9	4.2	8.7	21.3
	TS	1.4	4.8	11.2	18.1
	Platt (PS)	1.5	4.5	10.8	17.6
<i>Soft (ours)</i>	MCTS	1.5	2.9	4.1	7.3
	SLTS	1.3	2.4	3.8	6.2
	IR-Soft	1.1	2.3	3.5	6.7

404 G Extended Results: Stratified ECE on CIFAR-10H

405 H Implementation Details

406 **CIFAR-10H models.** *ResNet-50*: ImageNet-1k weights (ResNet50_Weights.IMAGENET1K_V1);
 407 FC layer replaced with 2048×10 linear; AdamW, lr= 10^{-4} , weight decay 10^{-4} , batch 128, cosine
 408 annealing, 30 epochs, 224×224 . *ViT-B/16*: ImageNet-21k weights; classification head replaced;
 409 AdamW, lr= 5×10^{-5} , weight decay 10^{-4} , batch 64, cosine annealing, 20 epochs, 224×224 .

410 **ChaosNLI models.** *RoBERTa-Large*: roberta-large-mnli from HuggingFace, pre-fine-
 411 tuned on MultiNLI [Liu et al., 2019]; no additional fine-tuning. *DeBERTa-v3-base*:
 412 cross-encoder/nli-deberta-v3-base from HuggingFace [He et al., 2023]; no additional fine-
 413 tuning. Both models use the combined ChaosNLI SNLI+MNLI subset ($n = 3,113$), split 50/50 into
 414 calibration and test.

415 **ISIC 2019 models.** *EfficientNet-B4*: ImageNet-1k weights; stratified 70/15/15 split; class-weighted
416 cross-entropy (NV: $\approx 67\%$ of training images); AdamW, $\text{lr} = 3 \times 10^{-4}$, weight decay 10^{-4} , 2-epoch
417 linear warm-up + cosine decay, 20 epochs, 380×380 . *ViT-S/16*: ImageNet-21k weights; same split
418 and loss weighting; AdamW, $\text{lr} = 10^{-4}$, weight decay 10^{-4} , 2-epoch warm-up + cosine decay, 20
419 epochs, 224×224 .

420 **DermaMNIST models.** *ResNet-18*: ImageNet-1k weights, class-weighted cross-entropy, AdamW,
421 $\text{lr} = 10^{-4}$, 30 epochs. *ViT-S/16*: ImageNet-21k weights, same training protocol, 224×224 .

422 **Calibration optimizers.** TS, SLTS, MCTS: LBFGS, $\text{lr} = 0.1$, 500 iterations, tolerance 10^{-9}
423 (gradient), 10^{-11} (loss). Platt/VS: Adam, $\text{lr} = 0.01/0.05$, weight decay 10^{-4} , 2000 steps.

424 **ECE bins.** $B = 15$ equal-width bins in $[0, 1]$; bins with fewer than 1 example are excluded.

425 **MCTS.** $S = 50$ samples per calibration example, fixed seed 42. Sensitivity to S is negligible for
426 $S \geq 20$.

427 **Compute.** CIFAR-10H fine-tuning: ≈ 20 min on one NVIDIA A100. All calibration methods:
428 < 60 s on CPU after logit caching.